Survival-supervised Bayesian matrix factorization

Single-block model: how the October 2026 write-up differs from the March 2026 derivations, and whether survival still supervises $Y = LF^\top$ · prepared 29 Oct 2026

Documents compared: *MF_Derivations_UpdateAlgo_REVISED.pdf* (March 2026, "March") and *project_writeup_10_16_26.pdf* (16 Oct 2026, "October"). Scope: model and algorithm derivations of the single-block model only; applications, simulations and the multimodal model are not covered. Notation: $E[\cdot]$ is a posterior mean ($E[F]$ = posterior-mean loadings); $Z\sim = Y\,E[F]\,D^{-1}$ is the projection used in the October predictor; $\beta\sim$ is its Cox coefficient.

1. Bottom line

- **The model changed substantially.** In March the Cox predictor was $\eta = L\beta$ with L latent. In October it is $\eta = Z\sim\beta\sim$ with $Z\sim = Y\,E[F]\,D^{-1}$, a projection of the data onto the (column-normalized) loadings, so that new patients can be scored.
- **Survival no longer supervises the factorization in the evaluated single-block fit.** In March survival entered the L update directly (F only indirectly). In October, L does not appear in η at all, and the survival term for F has weight $\alpha_F = 0$, so survival determines only the Cox coefficients $\beta\sim$ (and, through the stopping rule, which iterate is returned). The fit is an unsupervised nonnegative EBMF interleaved with an empirical-Bayes Cox fit on projections.
- The write-up is candid about this (Summary, Sec. 1.2, 5.1, 5.3). It also replaces the "Taylor" Cox surrogate with an explicitly labelled working approximation and states that its monitoring statistic is not an ELBO.
- I re-derived and numerically checked the main updates in Appendix A and found no algebra errors. I did find a few mismatches between derivation and implementation, one likely sign-convention problem, and some overstated or unclear presentation (Section 5).

2. Key changes in model and algorithm

Aspect	March 2026	October 2026 (evaluated single-block fit)
Reconstruction	$Y = LF^\top + E$, $E_{ij} \sim N(0, \tau_j^{-1})$; no sign constraints stated.	Same, with $L, F \geq 0$ (point-exponential priors); Y standardized gene-by-gene within each training cohort.
Priors	Generic $g \in G$ fitted by EBNM; $\beta \sim N(0, \sigma^2)$, with σ^2 "subsumed" in EBNM.	Point-exponential for L and F (explicit truncated-normal posterior moments, A.4). $\beta\sim_k \sim N(0, \sigma_k^2)$, with closed-form $\sigma^2_{EB,k} = \max(0, x^2 - s^2)$ per factor (A.5).
Survival predictor	$\eta_i = l_i^\top \beta$ (latent loadings).	$\eta = Z\sim\beta\sim$, $Z\sim = Y\,E[F]\,D^{-1}$, D = diag of column norms of $E[F]$ (ridge 10^{-10}). $\beta\sim$ is per unit of normalized projection. Computable for new patients (frozen $E[F]$, D^*, $\beta\sim$).
Cox approximation	Second-order Taylor expansion; $W_{ii} = -\partial^2\ell/\partial\eta_i^2$ (diagonal of the Hessian; off-diagonal terms dropped without comment).	Exact Breslow score $u_i = \delta_i - a_i$ with Poisson-style curvature $w_i = \max(a_i, 10^{-6})$, $a_i = \exp(\eta_i)\cdot H_0(t_i)$ at the sweep-start predictor (expected events; H_0 = Breslow baseline). Stated explicitly to be "not the second-order Taylor expansion" (A.2).
Loading uncertainty in survival term	Error-in-variables: A_β uses $E[l^2] = \mathrm{Var}(l) + E[l]^2$, so uncertainty in L shrinks β.	Dropped. $Z\sim$ is a fixed covariate within each sweep (plug-in $E[F]$); A_β uses $Z\sim^2$ and no coefficient variance is propagated.
Survival weights	None.	$\alpha = 0.5$ on the survival term in the $\beta\sim$ update (same pseudo-observation, half the precision); α_F weights a survival term in the F update (evaluated: 0).
Sweep structure	Cox working quantities (z, W) once per outer iteration; then for each k: L, F, β; then τ.	Working quantities, $Z\sim$ and D fixed at sweep start; then for each k: $\beta\sim_k$, $L_{\cdot k}$, $F_{\cdot k}$; then τ.

Aspect	March 2026	October 2026 (evaluated single-block fit)
τ update	$\tau_j = n / \Sigma_i E[R_{ij}^2]$, with variance-inflated expected squared residual.	Same formula (A.8), with Q_j floored at $n \cdot 10^{-8}$.
Objective and stopping	Full ELBO; convergence on max change in E[L], E[β].	Monitoring statistic J with unequal weights $(1-\alpha, \alpha p)$, survival term evaluated at the start-of-sweep predictor, and approximate KL for L. Declared not an ELBO; ascent not guaranteed. Stop when relative change in $J < 10^{-5}$ after sweep 5.
Identifiability	SVD rotation of E[L] every 10 iterations (second moments reset).	Scale handled through D in the predictor.
Honesty about target vs algorithm	Implementation notes (second-moment reset, one Taylor step per outer loop, single-observation EBNM for β).	Table 12 compares the joint target with the evaluated algorithm component by component.

3. Does survival supervise the factorization $Y = LF^\top$?

The table lists, for each update, whether a survival term enters it. "Joint target" is the objective in the October Appendix A.1; the evaluated algorithm is what was run (K = 7, α = 0.5, $\alpha_F = 0$).

Update	March 2026	October: evaluated fit	October: joint target
L	**Yes, directly.** $A_{ik} = \Sigma_j \tau_j f^2_{jk} + W_{ii}\beta^2_k$, $B_{ik} = \Sigma_j \tau_j R_{ij} E[f_{jk}] + W_{ii} z_i^{(-k)} E[\beta_k]$ ("differs from standard EBMF").	**No.** L is not in η; reconstruction terms only (A.6).	**No.** Same.
F	No direct term (reconstruction only); influenced only indirectly through L.	**No.** $\alpha_F = 0$ so Eq. 18 reduces to the reconstruction update. Z~ is built from E[F] but is held fixed within the sweep, so F is treated as data in the coefficient update.	Yes, through Z~; the term is coupled across features and requires d_k held fixed (Eq. 17).
β / β~	Survival term only (EBNM).	Survival term only, tempered by α = 0.5.	Survival term only.
Anything else	—	Survival can change the returned iterate through the stopping rule (J).	—

- **March:** survival guided the factorization through L. The L update combined the genomics and survival likelihoods, and the β update carried an error-in-variables correction.
- **October, evaluated fit:** no update of L or F contains a survival term. β/β~ has no feedback path to the factorization, so E[L] and E[F] are those of an unsupervised nonnegative EBMF (up to the stopping point). Any advantage over a two-step pipeline comes from how the programs are fitted and the coefficients shrunk, not from survival-guided discovery; the write-up says exactly this.
- **October, $\alpha_F > 0$ option:** Eq. 18 adds a survival term to the F update, but it omits the normalizer d_k and the feature-coupling term and updates all features of a factor at once, so it is not the F update of the joint target. L is never supervised directly in the October design, even in the target.
- **Why it matters:** the "supervised" label describes the target, not the evaluated fit. Claims about survival-guided programs or recovery of low-prognostic-variance programs are not tested by the $\alpha_F = 0$ configuration. The write-up lists survival-guided loadings as the first aim of continuing work (Sec. 5.3).

4. Joint target versus evaluated algorithm

The write-up distinguishes the objective that motivates the method (the **joint target**, Eq. 7) from what was actually run (the **evaluated algorithm**, A.1 and Table 12). **The evaluated algorithm is the implemented one**; all reported single-block results come from it. It is not coordinate ascent on the joint target.

Joint target: $\mathcal{L}(q) = E_q\,\ell_Y + E_q\,\ell_C(\eta) - \Sigma KL(L) - \Sigma KL(F) - \Sigma KL(\beta\sim)$, with $\eta = \Sigma_k\,\beta\sim_k\, y^\top u_k(F)$, $u_k(F) = F_{\cdot k}/\|F_{\cdot k}\|$, F random inside the projection, and ℓ_C the exact Breslow partial likelihood.

Component	Joint target (Eq. 7)	Evaluated algorithm (implemented)
Survival term	$E_q\,\ell_C(\eta)$, exact	Expected working quadratic, $w_i = \max(a_i, 10^{-6})$, fixed at the sweep-start predictor (A.2)
Predictor	Random F; normalizer is the column norm of F	$Z\sim = Y\,E[F]\,D^{-1}$ from posterior means, fixed within a sweep; D has a ridge; prediction uses stored D*
β~ update	Not conjugate under the exact target	Plug-in with $Z\sim^2$, no projection variance, survival weight $\alpha = 0.5$
L update	Reconstruction only	Same
F update	Reconstruction plus coupled survival term (needs d_k fixed)	Reconstruction only ($\alpha_F = 0$)
τ	Maximum likelihood	Maximum likelihood with a floor
Objective	$\mathcal{L}(q)$	Monitoring statistic J, used only for stopping
β~ uncertainty	Posterior variance of β~	Stored variances are zero

- J differs from $\mathcal{L}$ in four ways: unequal weights on the reconstruction and survival components, a survival term evaluated at the start-of-sweep predictor, the working curvature in place of the exact one, and an approximate KL for L. It is not an ELBO and is not guaranteed to increase.
- Only the reconstruction updates (L, F, τ) are exact coordinate updates, of an unsupervised EBMF. The coefficient update is an empirical-Bayes Cox step on fixed projections.
- **Effect of the mapping $Z\sim = Y\,E[F]\,D^{-1}$.** Survival becomes a function of the data and the loadings, not of L. Y is then used twice (as the reconstruction response and as the Cox covariate), so the joint is $p(Y \mid L, F) \cdot p(t, \delta \mid Y, F, \beta\sim)$, a partial-likelihood target rather than a generative model. D removes the arbitrary scale of each loading column, so β~ is per unit of normalized projection. New patients can be scored without inferring L. Training projections are in-sample, since E[F] was fitted to the same Y.
- **L update and prior.** L keeps a point-exponential prior fitted by marginal likelihood across the n pseudo-observations (prior refit first, then the posterior under the refit prior). Its update has the same form as March but without the survival terms: $A_{ik} = \Sigma_j\,\tau_j\,E[f_{jk}^2]$ and $B_{ik} = \Sigma_j\,\tau_j\,R_{ij}^{(-k)}\,E[f_{jk}]$ (A.6). March added $W_{ii}\beta_k^2$ to A and $W_{ii}z_i^{(-k)}E[\beta_k]$ to B.

5. Checks of the October derivations (single block)

Verified. I re-derived by hand, and numerically where noted, the Cox score, Hessian and a_i (finite differences); the point-exponential posterior weights and moments (numerical integration); the coefficient update and its tempering (A.5, including the normal-prior EB formulas and the σ^2_{EB} maximizer, checked numerically); the L, F and τ updates; the KL identity and the κ_L discrepancy in A.9. No algebra errors found.

Issues, in rough order of importance (items 1–2 should be confirmed against the code):

1. **Sign convention (Sec. 1.2.1, A.10).** For the evaluated fit "higher η denotes longer survival" and the risk score is −η; Program 7 (described as adverse) has $\beta\sim = -0.040$ on the η scale. Under Eq. 6 (hazard $\propto e^{\eta}$) a fit that maximizes the partial likelihood should give an adverse program a positive coefficient. This looks like a sign bug or an undocumented convention, not a free orientation to be fixed from training data. (The multimodal model uses higher η = higher risk.)
2. **Coefficient variance (A.5 vs A.10).** A.5 gives $\mathrm{Var}(\beta\sim) = \sigma^2_{EB}s^2/(\sigma^2_{EB} + s^2) > 0$ whenever $\beta\sim \neq 0$, yet the stored variances are zero. This affects the variance term in J, the classification rule (which falls back to $|\beta\sim| > 0.001$), and $E[\beta\sim^2]$ in Eqs. 17–18 if $\alpha_F > 0$.
3. **One-observation EB for the coefficient prior (A.5).** $\sigma^2_{EB} = \max(0, x^2 - s^2)$ is nonzero whenever $|x| > s$, i.e. about 32% of null factors (about 16% at $\alpha = 0.5$, where s is inflated by $\sqrt{2}$). This is consistent with the 22–35% false survival-active rates reported in Sec. 3.1. March already flagged

single-observation EBNM as a weakness; the closed form keeps it. Pooling across factors or a hyper-prior on σ^2 would help.

4. **Nonnegative L, F on standardized data.** With L, F ≥ 0, the model mean $LF^\top$ is ≥ 0, but each gene is standardized to mean 0, so about half the entries are negative and cannot be represented. The multimodal simulation section itself notes the nonnegative model needs nonnegative data. Check that this is intended.
5. **No safeguard on the Cox surrogate.** The working quadratic is not a global lower bound (in a toy check it exceeded the true log partial likelihood in 136 of 2,000 random steps), and there is no damping. If a guarantee is wanted, $\mathrm{diag}(D_i/2)$, with D_i the number of deaths up to t_i, majorizes the negative Hessian (Böhning-type bound; verified numerically, not proved), at the cost of being conservative.
6. **Train/predict mismatch (minor).** Training projections use E[F] fitted to the same Y (in-sample), while new-patient projections do not. Prediction also pairs the final E[F] with D* and β~ from the start of the final sweep; only the norm mismatch ($< 2\times10^{-3}$) is disclosed.

Presentation.

- Symbol collisions: a_i is the expected number of events (A.2), then $Z\sim_{ik}$ (A.5), then y_{ij}/d_k (A.7); d_a (ties) vs d_k (norms); H_i, H, $\mathcal{H}$, H_0; S (monitor, S_+, Σ Pr); T; G; V.
- A.1 says “two features” make the algorithm differ from coordinate ascent on the target; α-weighting, the floors on w and A, the ridge in D and the approximate KL also differ (they appear later, in A.5 and Table 12).
- The single-block PVE (Sec. 1.3) is called proportion of variance explained but ignores non-orthogonality of the factors; the multimodal table warns that shares are not additive, the single-block definition does not.
- The weights (1−α) on reconstruction and αp on survival in J are unexplained and do not match the weights used in the updates.

6. Suggested direction

- **Prefer η = Lβ for the model, with fold-in for prediction.** It lets survival guide L (and F through L) and gives one coherent model with a proper error-in-variables term. A new patient is scored by running the L update with F and the prior fixed, then $\eta_{new} = l_{new}{}^\top E[\beta]$. This is my judgment, not a tested result.
- If the projection is kept, make supervision real ($\alpha_F > 0$ with the correct coupled update), propagate loading uncertainty through $E[Z\sim^2]$, and resolve the sign and coefficient-variance issues first.
- Compare the variants on simulations in which survival is generated from L. The write-up’s own simulation generates survival from L but predicts both arms from projections (Sec. 2.2), which favors the projection.

7. Scope and caveats

Based only on the two PDFs; I did not see the code, so items 1–2 in Section 5 are inferences from the text. I did not audit the March document line by line. Numerical checks used small random examples (Cox derivatives, posterior moments, coefficient formulas, scale equivariance). The multimodal model, simulations and applications are out of scope.